VERDANT AI

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VERDANT AI

Green Building Intelligence & Digital Twin Platform

- Problem: Buildings waste electricity by cooling and lighting empty or underutilized classrooms.
- Solution: An AI-powered platform that monitors occupancy, optimizes classroom allocation, and reduces unnecessary energy consumption.
- Core Innovation: RoomWise™ Session Consolidation – shifting classes into fewer suitable rooms so unused spaces can remain powered off.

Technical Approach

- Computer Vision: OpenCV and ONNX-based person detection for real-time occupancy monitoring.
- Backend: Python, FastAPI, and SQLite for data processing and storage.
- AI Optimization: RoomWise™ uses heuristic algorithms and Mixed-Integer Linear Programming (MILP) to optimize classroom allocation.
- Digital Twin: Interactive building visualization with energy monitoring and thermal simulation.
- AI Copilot: Groq Cloud LPU and Ollama for natural-language facility insights.

Privacy: Video frames are processed temporarily in memory, without being saved as recordings.

Workflow

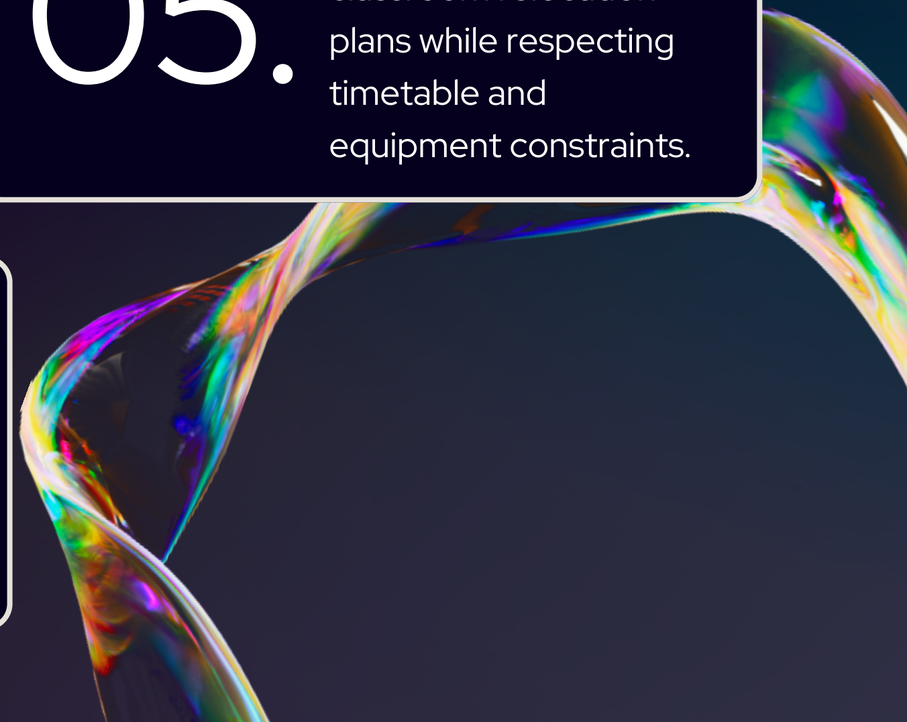
01. Occupancy Detection: Detect and stabilize classroom headcounts using camera input.

02. Data Collection: Combine occupancy readings with classroom schedules, capacities, and energy data.

03. Energy Analysis: Identify underutilized rooms and unnecessary energy consumption.

04. Approval & Monitoring: Facility managers review recommendations and monitor projected savings through the digital twin.

05. Schedule Optimization: Generate suitable classroom relocation plans while respecting timetable and equipment constraints.



Impact and Benefits

- 17% modeled daily energy reduction — equivalent to 88.3 kWh/day in the documented Academic Block A scenario.
- Reduced Carbon Emissions: Lower electricity consumption reduces associated emissions.
- Improved Space Utilization: Consolidates classes into suitable rooms.
- Privacy Protection: No video frames are stored as recordings.
- Predictive Maintenance: Identifies potential equipment faults and energy inefficiencies.
- Smart Decision-Making: Provides real-time dashboards, optimization recommendations, and sustainability reports.

References

- ASHRAE — Handbook—Fundamentals (building energy and thermal modeling).
- ASHRAE — Guideline 14-2014 (measurement and verification of energy savings).
- ASHRAE — Standard 55 (thermal comfort).
- ASHRAE — Standard 62.1 (ventilation and indoor air quality).
- ISO 52120-1:2021 — Building automation and energy efficiency.
- Efficiency Valuation Organization (EVO) — International Performance Measurement and Verification Protocol (IPMVP).
- Brick Schema v1.3 — Semantic modeling of buildings.
- Project documentation: VERDANT AI Technical Dossier and README.



The End